Environmental Economics: An Introduction Barry Field, 2024 Release (Chapter 8)

Questions for Further Discussion

1. Over the last two years, emission abatement costs in industry X have been about $1 million per year. A new regulation will lead to abatement costs of $1.8 million per year. Does this mean that the regulation will cause increased abatement costs of $800,000 per year? Explain.
2. In order to protect the quality of its nearby water resources, a community places a restriction on any housing development closer than 100 feet to a wetland. How might you estimate the social costs of this regulation?
3. Consider a small home project or a simple activity to help the climate that you, and maybe some friends, could do. Itemize and estimate the costs, including opportunity costs, of the project or activity if done by one person. Adjust your estimates if it expands to 10 people doing the activity or project. Then adjust your estimates if the activity or project expands for 5 years into the future. Discuss.
4. A tax on gasoline is proposed in order to raise money for the pollution-control activities of several public agencies. The tax will be 10¢ per gallon, and last year 10.3 million gallons of gasoline were used by motorists (this is strictly an illustrative number). Does this mean that we can anticipate $1,030,000 in revenues from this tax? Explain and use a graph to answer this question.
5. Most industries are composed of firms that, though perhaps producing roughly the same thing, are very different; some are large and some small; some are profitable and others not; some are located in one part of the country and some in others; some perhaps have undertaken a certain amount of voluntary emissions reductions and some have not. How does this complicate the job of estimating the total social costs of pollution-control regulations?